Théories des Langages

Exercices d'application :

Exo1 :

1- $10001^3 = 1000111$

$1(101)^2 = 1\,101\,101$

$101^R = 101$ R : miroir

$(10a)^R = 001$

2- $101011^3 = 101\,(011)^3 = 101\,011\,011\,011$

$101^2 = 101\,101$

$101^R = 101$

$(101\,111)^R = 111\,101$

$|101| = 1$

$|101\,101\,111| = 3$

Prop :

L_1, L_2, L_3 :

$L_1 \cdot (L_2 \cup L_3) = L_1 \cdot L_2 \cup L_1 \cdot L_3$: distributif

التوزيع

$(L_2 \cup L_3) \cdot L_1 = L_2 \cdot L_1 \cup L_3 \cdot L_1$

$L_1 \cdot L_2 \neq L_2 \cdot L_1$: ChaymaAymen ≠ AymenChayma

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Ex2:

$L_1 = \{\varepsilon, aa\}$, $L_2 = \{a^i b^j \mid i,j \geq 0\}$ et $L_3 = \{ab, b\}$

Calculer :

- $L_1 . L_2 = \{\varepsilon, aa\} . \{a^i b^j \mid i,j \geq 0\}$

$= (\{\varepsilon\} \cup \{aa\}) . L_2$

$= \{\varepsilon\} . L_2 \cup \{aa\} . L_2$

$= L_2 \cup \{aa\} . L_2$

$= L_2 \cup \{aaa^i b^j \mid i,j \geq 0\}$

$= L_2 \cup \{a^k b^j \mid j \geq 0, k \geq 2\}$

$= L_2 \Rightarrow car \{a^k b^j \mid j \geq 0 \text{ et } k \geq 2\} \subset L_2$

Si $A \cap B$ alors $A \cup B = B$

- $L_1 . L_3 = \{\varepsilon, aa\} . \{ab, b\}$

$= \{ab, b, aaab, aab\}$

- $L_1 \cup L_2 = L_2 \Rightarrow car\ L_1 \subset L_2$

$\Rightarrow$

- $L_2 \cap L_3 = L_3$ car $L_3 \subset L_2$ donc $L_2 \cap L_3 = L_3$

Si $A \subset B$ alors $A \cap B = A$

- $L_1^{10} = \{v_1 . v_2 \dots v_{10} \mid v_i = \varepsilon \text{ ou } v_i = aa \} \ 0 \leq i \leq 10\}$
- $L_1^* = L_1^{10} = \{(aa)^i / 0 \leq i \leq 10\}$
- $L_1^+ =$
- $L_2^R =$
- $L_1^* = \{\varepsilon, aa\} = (\{\varepsilon\} \cup \{aa\})^*$

$= (\{\varepsilon\}^* \mid \{aa\}^*)^*$

$= (\{aa\}^*)^* = \{aa\}^* = \{(aa)^i / i \geq 0\} = L_1^*$

$L^* = (L^*)^*$

$L^* = L^* . L^*$

$(L_1 + L_2)^* =$

$(L_1^* . L_2^*) = (L_1^* + L_2^*)$

$L^* = L^0 \cup L^1 \cup L^2 \ldots \cup L^m$

$L^0 = \{\varepsilon\}$

$L_1^+ = L^1 \cup L^2 \ldots \cup L^m$

$L^+ = L . L^*$

- si le mot vide $\in$ à L donc $L^+ = L^*$ (ε)

$L_1^+ = L_1 . L_1^*$

$= (\{\varepsilon\} \cup \{aa\}) . L_1^*$

$= \{\varepsilon\} . L^* , \cup \{aa\} . L_1^*$

$= L_1^* \cup \{aa^i / i \geq 1\}$

$L_1^+ = L_1^*$

$L_2^R = \{b^j a^i / i, j \geq 0\}$

EXO3 :

2) Longueur = 3

3) $\Sigma = \{+, *, -, /, (\ ), 0, 1, 2, 3, 4, \ldots, 9\}$

$\Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, +, -, *, /, (,)\}$

4) Fermeture

EX4 :

1) $L = \{1, 2, 3, 4, 5, 6, 7, 8, 9\} . \Sigma^* \cup \{0\} \Rightarrow \Sigma = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$

$\{+, -, \varepsilon\} . \{.\} . \Sigma^* \cup \{0\}$

1) $L_z = \{+, -, \varepsilon\} \cdot (\{1,2,3,\dots,9\} \cdot \varepsilon^* \cup \{0\})$

2) Nombre décimal = entier . entier

$L_z \cdot \{\cdot\} \cdot L_z$

$\Sigma = \{0,1,2,3,4 \dots, 9, -, +, \cdot\}$

$L_{decimal} = \{+, -, \varepsilon\} \cdot (\{1,2,3,4,5,6,7,8,9\} \cdot \{0,1,2,3,4,5,6,7,8,9\}^* \cup \{0\}) \cdot \{\cdot\} \cdot \{0,1,2,3,4,5,6,7,8,9\}^*$

partie entière

partie décimal

Ex :

$\Sigma = \{a, b, c, d\}$

$L = \{a\} \cdot \Sigma^*$

$L = \Sigma^* \cdot \{a\}$

$L = \{a\} \cdot \Sigma^* \cdot \{b\}$

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